Reliable evaluation of image-to-patient MES classification

How do prediction aggregation and the derived reference affect patient-level conclusions?

A IMAGE PREDICTION → PATIENT SUMMARY

LIMUC images

Group all images by patient identifier

Image classifiers

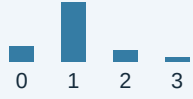
ViT-B/16 or Swin-B

Class-weighted CE + optional ordinal term

Image-label supervision

Image outputs

MES probabilities for each image (0–3)



Patient prediction

Aggregate within each patient

Compare six candidate rules

Inner-selected rule:

mean probability → argmax

B DERIVED REFERENCE AND EVALUATION

Released image labels

MES 0–3 for the same patient-image bag

Derived patient reference

Primary: modal-low

Sensitivity: modal-high, floor / ceiling of median, maximum

Reference construction is separate from prediction aggregation.

vs.

Patient-level evaluation

Ordinal agreement: QWK, MAE

Classes: macro-F1, accuracy

Probability quality: Brier, ECE

C PATIENT-GROUPED NESTED VALIDATION

5 outer folds

Keep each patient's images together; hold out one fold

4-fold inner selection

Select loss, aggregation, and model family

Refit → outer evaluation

Evaluate held-out patients; pool predictions across outer folds

Post hoc diagnostics: cross aggregation × reference choices and inspect probability quality; no reselection.

Target: a derived patient-image-bag summary, not an independently adjudicated examination-level MES.